1. Problem Title & Source

Title: **Verbal Kho-Kho Digit Communication**

Source: **TCS NQT Previous Coding Questions**

2. Problem Statement (Clear Version)

Alice and her friends are playing a communication game called **Verbal Kho-Kho**.

There are **N friends** sitting in a line.

The process works like this:

1. Alice writes a **single digit** on a paper and gives it to the **first friend**.
2. Each friend tries to **communicate the digit to the next friend without speaking**.
3. The digit passes from:

Friend1 → Friend2 → Friend3 → … → FriendN

4. Finally, the **last friend writes the digit** they understood and gives it back to Alice.

Alice then checks whether the communication worked correctly.

You are given:

- N → number of friends
- D[i] → digit understood by each friend

Important rule:

The correct digit should be the digit understood by the first friend.

Your task is to **count how many friends did NOT understand the digit correctly**.

3. Examples

Example 1

Input

3

4 4 4

Output

0

Explanation

All friends understood the same digit:

4 → 4 → 4

No communication error occurred.

Example 2

Input

5

1 2 3 2 2

Output

4

Explanation

Correct digit =

1

Check each friend:

Friend	Digit	Correct?
1	1	Yes
2	2	No
3	3	No
4	2	No
5	2	No

Number of incorrect friends:

4

4. Constraints

1 ≤ N ≤ 100

0 ≤ D[i] ≤ 9

Observations

- Digits range from **0–9**
- Simple array comparison problem

5. Core Concept (Pattern)

Array Traversal + Counting

Key idea:

Compare every element with the first element.

6. Thought Process

Step 1 — Identify Correct Digit

The correct digit is:

D[0]

Because it was originally given by Alice.

Step 2 — Compare All Friends

For each friend:

if digit ≠ correctDigit

```
incorrectCount++
```

Step 3 — Return Incorrect Count

Total incorrect friends represent the **communication failure count**.

7. Visualization

Example

D = [1,2,3,2,2]

Correct digit

1

Comparison

Friend	Digit	Match?
1	1	Yes
2	2	No
3	3	No
4	2	No
5	2	No

Result

4

8. Pseudocode

```
read N
read array D

correctDigit = D[0]

count = 0

for i from 0 to N-1

   if D[i] != correctDigit
      count++

print count
```

9. Code Implementation

Python

```
n = int(input())
arr = list(map(int,input().split()))

correct = arr[0]

count = 0

for x in arr:
    if x != correct:
        count += 1

print(count)
```

Java

```
import java.util.*;

class Main {

    public static void main(String[] args){

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int[] arr = new int[n];

        for(int i=0;i<n;i++)
            arr[i] = sc.nextInt();

        int correct = arr[0];

        int count = 0;

        for(int x : arr){

            if(x != correct)
```

```
            count++;
        }

        System.out.println(count);
    }
}
```

10. Time & Space Complexity

Time Complexity

O(N)

We traverse the array once.

Space Complexity

O(1)

Only a counter variable used.

11. Common Mistakes

1. Starting loop from index 1 but forgetting to check the first element correctly
2. Misinterpreting the correct digit
3. Counting correct instead of incorrect
4. Using nested loops unnecessarily

12. Detailed Dry Run

Input

5

1 2 3 2 2

Correct digit

1

Traversal

Index	Digit	Incorrect Count
0	1	0
1	2	1
2	3	2
3	2	3
4	2	4

Output

4

13. Real-Life Applications

This type of pattern appears in:

- Error detection systems
- Communication verification
- Data consistency checks
- Sensor signal validation

14. Common Traps

1. Forgetting that the **first element is always correct**
2. Misinterpreting the problem as **majority element**
3. Using extra arrays unnecessarily
4. Incorrect input reading

15. Related Problems

Array comparison problems:

LC 169 Majority Element

LC 136 Single Number

LC 217 Contains Duplicate

16. Alternate Approaches

Approach	Complexity	Notes
Direct traversal	O(N)	optimal
Frequency map	O(N)	unnecessary

17. Why This Solution Works

Since the **first friend received the correct digit**, every other friend's value must be compared against it.

Any mismatch indicates **incorrect communication**.

Thus a single pass through the array is sufficient.

18. Variations / Follow-Up Questions

Possible interview variations:

1. Count friends who **understood correctly**
2. Find the **most common digit communicated**

3. Detect **communication chains**
4. Find **first incorrect friend**